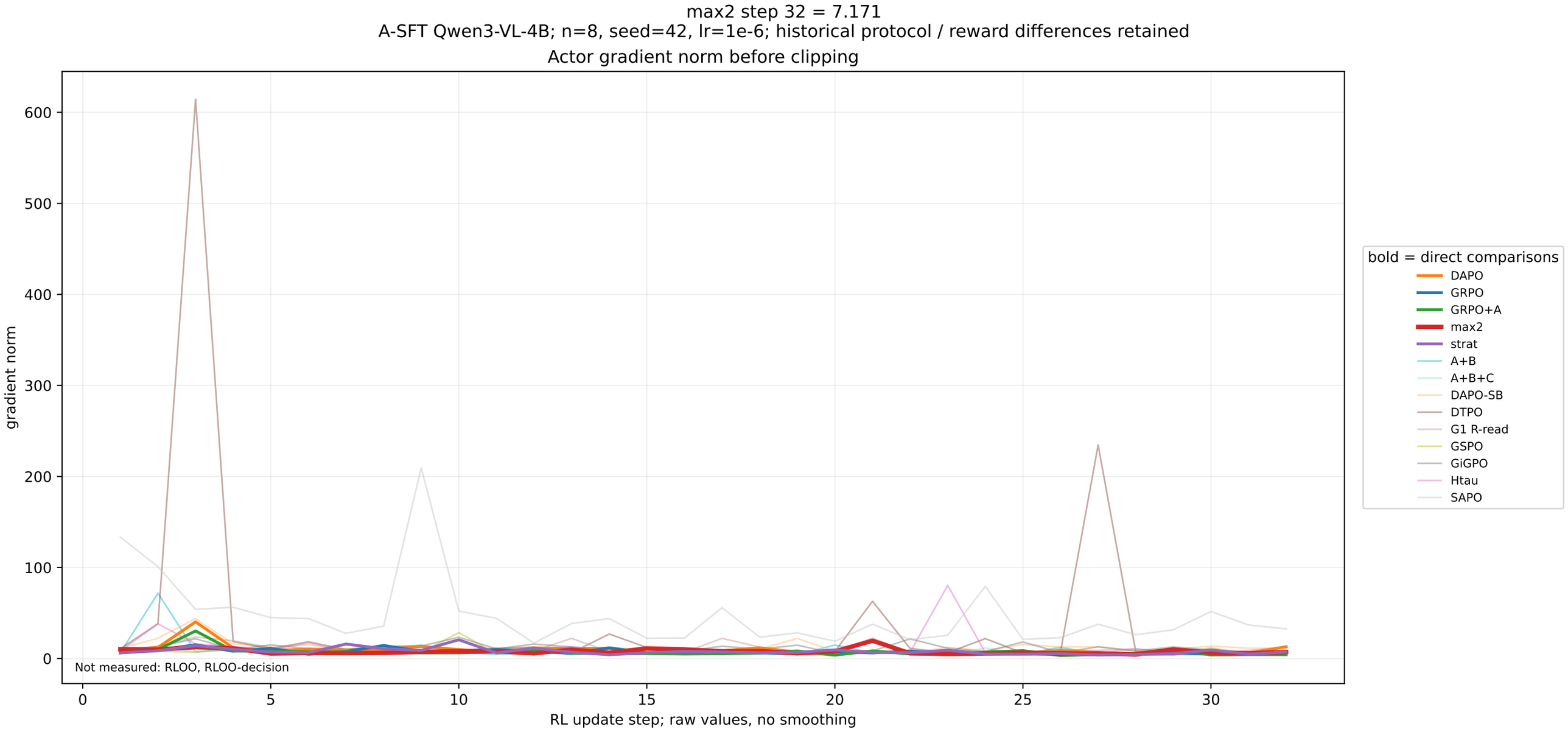
max2 step 32 = 7.171
A-SFT Qwen3-VL-4B; n=8, seed=42, lr=1e-6; historical protocol / reward differences retained
Actor gradient norm before clipping
gradient norm
RL update step; raw values, no smoothing
Not measured: RLOO, RLOO-decision
bold = direct comparisons
DAPO
GRPO
GRPO+A
max2
strat
A+B
A+B+C
DAPO-SB
DTPO
G1 R-read
GSPO
GiGPO
Htau
SAPO